United States Patent	12404099
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Austrheim; Trond et al.

Rescue system and methods for retrieving a malfunctioning vehicle from a rail system

Abstract

A rescue system for retrieving a malfunctioning vehicle from a rail system of an automated storage and retrieval system includes a first rescue vehicle and a second rescue vehicle. The rail system includes a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction perpendicular to the X-direction, and a plurality of remotely operated vehicles configured to move in the X and Y-directions on the tracks of the rail system. The first rescue vehicle is configured to run on the tracks of the rail system. The first rescue vehicle is provided with a lifting device on one side of the vehicle, which faces for engagement in a first X-direction. The second rescue vehicle is configured to run on the tracks of the rail system. The second rescue vehicle is provided with a lifting device on an opposite side of the vehicle, which faces for engagement in a second X-direction opposite to the first X-direction. The first and second rescue vehicles are configured to work in tandem so that when one of the plurality of remotely operated vehicles malfunctions, the first and second rescue vehicles can position themselves on the rail system on opposite sides of the malfunctioning vehicle to engage their respective lifting devices with the opposite sides of the malfunctioning vehicle. The first and second rescue vehicles operate their lifting devices simultaneously so as to lift the malfunctioning vehicle off the rail system and transport the malfunctioning vehicle.

Inventors:	**Austrheim; Trond (Etne, NO), Gjerdevik; Øystein (Skjold, NO)**
Applicant:	**Autostore Technology AS** (Nedre Vats, NO)
Family ID:	**73059944**
Assignee:	**AutoStore Technology AS (Nedre Vats, NO)**
Appl. No.:	**17/770563**
Filed (or PCT Filed):	**November 04, 2020**
PCT No.:	**PCT/EP2020/080929**
PCT Pub. No.:	**WO2021/094161**

PCT Pub. Date: May 20, 2021

Prior Publication Data

Document Identifier	**Publication Date**
US 20220388773 A1	Dec. 08, 2022

Foreign Application Priority Data

NO	20191338	Nov. 12, 2019

Publication Classification

Int. Cl.: **B65G1/04** (20060101); **B65G1/06** (20060101)

U.S. Cl.:

CPC **B65G1/0464** (20130101); **B65G1/0478** (20130101); **B65G1/065** (20130101); B65G2207/40 (20130101)

Field of Classification Search

CPC: B65G (1/0464); B65G (1/0478); B65G (1/065); B65G (2207/40); B65G (2201/0258); B65G (2207/30); B65G (1/04); B65G (1/0492); B65G (1/06); B65G (1/0414)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2018/0194571	12/2017	Fryer	N/A	B65G 1/0478
2019/0054932	12/2018	Stadie	N/A	B61B 5/02

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109748022	12/2018	CN	N/A
2539562	12/2015	GB	N/A
2016141323	12/2015	JP	N/A
2017509564	12/2016	JP	N/A
345931	12/2020	NO	N/A
2015/140216	12/2014	WO	N/A
2015/193278	12/2014	WO	N/A
2017/148939	12/2016	WO	N/A
2018/146304	12/2017	WO	N/A
2019/206672	12/2018	WO	N/A

OTHER PUBLICATIONS

International Search Report issued in PCT/EP2020/080929 on Feb. 8, 2021 (4 pages). cited by applicant

Written Opinion of the International Searching Authority issued in PCT/EP2020/080929 on Feb. 8,

2021 (11 pages). cited by applicant
Written Opinion of the International Preliminary Examining Authority issued in PCT/EP2020/080929 mailed on Sep. 23, 2021 (12 pages). cited by applicant
Norwegian Search Report issued in NO 20191338 mailed on Jun. 10, 2020 (2 pages). cited by applicant
Mari Uchida, Notice of Reasons for Rejection for Japanese Patent Application No. 2022526366, dated Oct. 25, 2024, 15 pages, pub. by the JPO. cited by applicant
Mari Uchida, Notice of Reasons for Rejection for Japanese Patent Application No. 2022526366, dated Mar. 12, 2025, 4 pages, pub. by the JPO, Tokyo, Japan. cited by applicant
Yang, Young June, Office Action in KR1020227019246, mailed Jun. 16, 2025, 17 pages, Korean Intellectual Property Office, Daejeon, Korea. cited by applicant

---

Primary Examiner: Randazzo; Thomas

Attorney, Agent or Firm: Baker Botts L.L.P.

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to an automated storage and retrieval system for storage and retrieval of containers, and in particular to a rescue system for retrieving a malfunctioning vehicle from a rail system, a method of retrieving a malfunctioning container handling vehicle from a rail system and a rescue vehicle for use in said systems and method.

BACKGROUND AND PRIOR ART

(2) FIG. **1** discloses a typical prior art automated storage and retrieval system **1** with a framework structure **100** and FIGS. **2** and **3** discloses two different prior art container handling vehicles **201**,**301** suitable for operating on such a system **1**.

(3) The framework structure **100** comprises upright members **102**, horizontal members **103** and a storage volume comprising storage columns **105** arranged in rows between the upright members **102** and the horizontal members **103**. In these storage columns **105** storage containers **106**, also known as bins, are stacked one on top of one another to form stacks **107**. The members **102**, **103** may typically be made of metal, e.g. extruded aluminum profiles.

(4) The framework structure **100** of the automated storage and retrieval system **1** comprises a rail system **108** arranged across the top of framework structure **100**, on which rail system **108** a plurality of container handling vehicles **201**,**301** are operated to raise storage containers **106** from, and lower storage containers **106** into, the storage columns **105**, and also to transport the storage containers **106** above the storage columns **105**. The rail system **108** comprises a first set of parallel rails **110** arranged to guide movement of the container handling vehicles **201**,**301** in a first direction X across the top of the frame structure **100**, and a second set of parallel rails **111** arranged perpendicular to the first set of rails **110** to guide movement of the container handling vehicles **201**,**301** in a second direction Y which is perpendicular to the first direction X. Containers **106** stored in the columns **105** are accessed by the container handling vehicles through access openings **112** in the rail system **108**. The container handling vehicles **201**,**301** can move laterally above the storage columns **105**, i.e. in a plane which is parallel to the horizontal X-Y plane.

(5) The upright members **102** of the framework structure **100** may be used to guide the storage containers during raising of the containers out from and lowering of the containers into the columns **105**. The stacks **107** of containers **106** are typically self-supportive.

(6) Each prior art container handling vehicle **201**,**301** comprises a vehicle body **201*a***,**301*a***, and

first and second sets of wheels **201***b*,**301***b*,**201***c*,**301***c* which enable the lateral movement of the container handling vehicles **201**,**301** in the X direction and in the Y direction, respectively. In FIGS. **2** and **3** two wheels in each set are fully visible. The first set of wheels **201***b*,**301***b* is arranged to engage with two adjacent rails of the first set **110** of rails, and the second set of wheels **201***c*,**301***c* is arranged to engage with two adjacent rails of the second set **111** of rails. At least one of set wheels **201***b*,**301***b*,**201***c*,**301***c* can be lifted and lowered, so that the first set of wheels **201***b*,**301***b* and/or the second set of wheels **201***c*,**301***c* can be engaged with the respective set of rails **110**, **111** at any one time.
(7) Each prior art container handling vehicle **201**,**301** also comprises a lifting device (not shown) for vertical transportation of storage containers **106**, e.g. raising a storage container **106** from, and lowering a storage container **106** into, a storage column **105**. The lifting device comprises one or more gripping/engaging devices which are adapted to engage a storage container **106**, and which gripping/engaging devices can be lowered from the vehicle **201**,**301** so that the position of the gripping/engaging devices with respect to the vehicle **201**,**301** can be adjusted in a third direction Z which is orthogonal the first direction X and the second direction Y. Parts of the gripping device of the container handling vehicle **301** is shown in in FIG. **3** and is indicated with reference number **304**. The gripping device of the container handling device **201** is located within the vehicle body **301***a* in FIG. **2**.
(8) Conventionally, and also for the purpose of this application, Z=1 identifies the uppermost layer of storage containers, i.e. the layer immediately below the rail system **108**, Z=2 the second layer below the rail system **108**, Z=3 the third layer etc. In the exemplary prior art disclosed in FIG. **1**, Z=8 identifies the lowermost, bottom layer of storage containers. Similarly, X=1 . . . n and Y=1 . . . n identifies the position of each storage column **105** in the horizontal plane. Consequently, as an example, and using the Cartesian coordinate system X, Y, Z indicated in FIG. **1**, the storage container identified as **106**′ in FIG. **1** can be said to occupy storage position X=10, Y=2, Z=3. The container handling vehicles **201**,**301** can be said to travel in layer Z=0, and each storage column **105** can be identified by its X and Y coordinates.
(9) The storage volume of the framework structure **100** has often been referred to as a grid **104**, where the possible storage positions within this grid is referred to as a storage cell. Each storage column may be identified by a position in an X- and Y-direction, while each storage cell may be identified by a container number in the X-, Y and Z-direction.
(10) Each prior art container handling vehicle **201**,**301** comprises a storage compartment or space for receiving and stowing a storage container **106** when transporting the storage container **106** across the rail system **108**. The storage space may comprise a cavity arranged centrally within the vehicle body **201***a* as shown in FIG. **2** and as described in e.g. WO2015/193278A1, the contents of which are incorporated herein by reference.
(11) FIG. **3** shows an alternative configuration of a container handling vehicle **301** with a cantilever construction. Such a vehicle is described in detail in e.g. NO317366, the contents of which are also incorporated herein by reference.
(12) The central cavity container handling vehicles **201** shown in FIG. **2** may have a footprint that covers an area with dimensions in the X and Y directions which is generally equal to the lateral extent of a storage column **105**, e.g. as is described in WO2015/193278A1, the contents of which are incorporated herein by reference. The term ‘lateral’ used herein may mean ‘horizontal’.
(13) Alternatively, the central cavity container handling vehicles **101** may have a footprint which is larger than the lateral area defined by a storage column **105**, e.g. as is disclosed in WO2014/090684A1.
(14) The rail system **108** typically comprises rails with grooves into which the wheels of the vehicles are inserted. Alternatively, the rails may comprise upwardly protruding elements, where the wheels of the vehicles comprise flanges to prevent derailing. These grooves and upwardly protruding elements are collectively known as tracks. Each rail may comprise one track, or each

rail may comprise two parallel tracks.
(15) WO2018146304, the contents of which are incorporated herein by reference, illustrates a typical configuration of rail system **108** comprising rails and parallel tracks in both X and Y directions.
(16) In the framework structure **100**, a majority of the columns **105** are storage columns **105**, i.e. columns **105** where storage containers **106** are stored in stacks **107**. However, some columns **105** may have other purposes. In FIG. **1**, columns **119** and **120** are such special-purpose columns used by the container handling vehicles **201**,**301** to drop off and/or pick up storage containers **106** so that they can be transported to an access station (not shown) where the storage containers **106** can be accessed from outside of the framework structure **100** or transferred out of or into the framework structure **100**. Within the art, such a location is normally referred to as a ‘port’ and the column in which the port is located may be referred to as a ‘port column’ **119**,**120**. The transportation to the access station may be in any direction, that is horizontal, tilted and/or vertical. For example, the storage containers **106** may be placed in a random or dedicated column **105** within the framework structure **100**, then picked up by any container handling vehicle and transported to a port column **119**,**120** for further transportation to an access station. Note that the term ‘tilted’ means transportation of storage containers **106** having a general transportation orientation somewhere between horizontal and vertical.
(17) In FIG. **1**, the first port column **119** may for example be a dedicated drop-off port column where the container handling vehicles **201**,**301** can drop off storage containers **106** to be transported to an access or a transfer station, and the second port column **120** may be a dedicated pick-up port column where the container handling vehicles **201**,**301** can pick up storage containers **106** that have been transported from an access or a transfer station.
(18) The access station may typically be a picking or a stocking station where product items are removed from or positioned into the storage containers **106**. In a picking or a stocking station, the storage containers **106** are normally not removed from the automated storage and retrieval system **1**, but are returned into the framework structure **100** again once accessed. A port can also be used for transferring storage containers to another storage facility (e.g. to another framework structure or to another automated storage and retrieval system), to a transport vehicle (e.g. a train or a lorry), or to a production facility.
(19) A conveyor system comprising conveyors is normally employed to transport the storage containers between the port columns **119**,**120** and the access station.
(20) If the port columns **119**,**120** and the access station are located at different levels, the conveyor system may comprise a lift device with a vertical component for transporting the storage containers **106** vertically between the port column **119**,**120** and the access station.
(21) The conveyor system may be arranged to transfer storage containers **106** between different framework structures, e.g. as is described in WO2014/075937A1, the contents of which are incorporated herein by reference.
(22) When a storage container **106** stored in one of the columns **105** disclosed in FIG. **1** is to be accessed, one of the container handling vehicles **201**,**301** is instructed to retrieve the target storage container **106** from its position and transport it to the drop-off port column **119**. This operation involves moving the container handling vehicle **201**,**301** to a location above the storage column **105** in which the target storage container **106** is positioned, retrieving the storage container **106** from the storage column **105** using the container handling vehicle's **201**,**301** lifting device (not shown), and transporting the storage container **106** to the drop-off port column **119**. If the target storage container **106** is located deep within a stack **107**, i.e. with one or a plurality of other storage containers **106** positioned above the target storage container **106**, the operation also involves temporarily moving the above-positioned storage containers prior to lifting the target storage container **106** from the storage column **105**. This step, which is sometimes referred to as “digging” within the art, may be performed with the same container handling vehicle that is subsequently

used for transporting the target storage container to the drop-off port column **119**, or with one or a plurality of other cooperating container handling vehicles. Alternatively, or in addition, the automated storage and retrieval system **1** may have container handling vehicles specifically dedicated to the task of temporarily removing storage containers from a storage column **105**. Once the target storage container **106** has been removed from the storage column **105**, the temporarily removed storage containers can be repositioned into the original storage column **105**. However, the removed storage containers may alternatively be relocated to other storage columns.

(23) When a storage container **106** is to be stored in one of the columns **105**, one of the container handling vehicles **201**,**301** is instructed to pick up the storage container **106** from the pick-up port column **120** and transport it to a location above the storage column **105** where it is to be stored. After any storage containers positioned at or above the target position within the storage column stack **107** have been removed, the container handling vehicle **201**,**301** positions the storage container **106** at the desired position. The removed storage containers may then be lowered back into the storage column **105**, or relocated to other storage columns.

(24) For monitoring and controlling the automated storage and retrieval system **1**, e.g. monitoring and controlling the location of respective storage containers **106** within the framework structure **100**, the content of each storage container **106**; and the movement of the container handling vehicles **201**,**301** so that a desired storage container **106** can be delivered to the desired location at the desired time without the container handling vehicles **201**,**301** colliding with each other, the automated storage and retrieval system **1** comprises a control system **500** which typically is computerized and which typically comprises a database for keeping track of the storage containers **106**.

(25) From prior art WO2015140216A1 it is known a robotic service device for use on a robotic picking system grid. The robotic service device is capable of driving to any location on the grid in order to perform maintenance operations or cleaning. Additionally, the service device may be used to rescue robotic load handling devices operational in the picking system. The robotic service device may comprise a releasable docking mechanism to enable it to dock and latch on to malfunctioning load handling devices. The service device may also be provided with cleaning means and camera means to enable the condition of the grid and other robotic devices to be monitored.

(26) It may be a problem with the prior art robotic service vehicles that the required lifting capacity of the single robotic service vehicle lifting a malfunctioning vehicle off the rail system is too high such that the robotic service vehicle is either not able to lift the malfunctioning vehicle and/or, if the robotic service vehicle is able to lift the malfunctioning vehicle off the rail system, it is unstable during transport requiring large counterweight(s) and or slow transportation speed/acceleration.

(27) The required lifting capacity may be even higher if the malfunction vehicle carries a heavy storage container. This may incur an even larger problem.

(28) It is an objective of the invention to solve the challenges of retrieving vehicles from a rail system.

SUMMARY OF THE INVENTION

(29) The present invention is set forth and characterized in the independent claims, while the dependent claims describe other characteristics of the invention. The invention provides for the possibility of rescuing container handling vehicles while the automated storage and retrieval system is in operation, i.e. while the remaining container handling vehicles are in operation on the rail system. The invention eliminates the need for a manned service vehicle improving HSE for the system. However, in the event of a major collision where vehicles are off track, it may be required that an operator enters the rail system.

(30) It is described a rescue system for retrieving a malfunctioning vehicle from a rail system of an automated storage and retrieval system, the rail system comprising a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction

perpendicular to the X-direction, a plurality of remotely operated vehicles configured to move in the X and Y-directions on the tracks of the rail system, a first rescue vehicle configured to run on the tracks of the rail system, the first rescue vehicle being provided with a lifting device on one side of the vehicle, the lifting device facing for engagement in a first X-direction; a second rescue vehicle configured to run on the tracks of the rail system, the second rescue vehicle being provided with a lifting device on an opposite side of the vehicle, the lifting device facing for engagement in a second X-direction opposite to the first; wherein the first and second rescue vehicles are configured to work in tandem so that when one of the plurality of remotely operated vehicles malfunctions, the first and second rescue vehicles can position themselves on the rail system on opposite sides of the malfunctioning vehicle to engage their respective lifting devices with the opposite sides of the malfunctioning vehicle and operate their lifting devices simultaneously so as to lift the malfunctioning vehicle off the rail system and transport the malfunctioning vehicle.
(31) The lifting device may comprise a vertical plate with a lip extending therefrom. Alternatively, other types of lifting devices may be used as long as they provide the required function of a horizontal part engageable with a malfunctioning vehicle such that the malfunctioning vehicle can be lifted off the rail system by vertical movement of the horizontal part relative the rail system. The lifting device may further comprise any necessary components required to lift and lower the lifting device relative the rail system, such as motor, any necessary guide or actuator for guiding the lifting device substantially vertical, connection to a power source for driving the motor etc. The motor, and possibly also the power source, may be designed with less lifting capacity than the prior art service vehicles which utilizes only one motor to lift a container handling vehicle off the grid.
(32) The container handling vehicle can be in the form of prior art container handling vehicles as the ones exemplified in FIGS. **2** and **3**A configured for receiving storage containers from below, or in the form of a container delivery vehicle configured for receiving storage containers from above.
(33) It is further described a rescue system for retrieving a malfunctioning vehicle from a rail system of an automated storage and retrieval system, the rail system comprising a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction perpendicular to the X-direction, the rails defining a plurality of grid cells, wherein a plurality of remotely operated vehicles are configured to move in the X and Y-directions on the tracks of the rail system, a first rescue vehicle comprising a first wheel base unit configured to run on the tracks of the rail system, the first wheel base unit providing a mobile platform corresponding in area to a single grid cell for a first vehicle rescue module mounted thereon, the first vehicle rescue module being orientated in a first direction of the rail system; a second rescue vehicle comprising a second wheel base unit configured to run on the tracks of the rail system, the second wheel base unit providing a mobile platform corresponding in area to a single grid cell for a second vehicle rescue module mounted thereon, the second vehicle rescue module being orientated in a second direction of the rail system opposite to the first direction; wherein the first and second rescue vehicles are configured to work in tandem to perform rescue operations on a malfunctioning vehicle of the plurality of remotely operated vehicles, the vehicle rescue modules being arranged to engage opposite sides of the malfunctioning vehicle and lift it off the rail system using a lifting device through being orientated in opposite facing directions.
(34) The vehicle rescue modules may comprise a lifting plate with a lip extending at a height above an upper surface of the wheel base unit. The height may be within 50 mm or it may be less or it may be more.
(35) The plurality of remotely operated vehicles may comprise wheel base units providing mobile platforms, each corresponding in area to a single grid cell of the rail system, for storage container lifting modules mounted thereon.
(36) A grid cell may be defined as the area, including the track width, delimited by a pair of tracks in X and Y direction around an access opening of the rail system.
(37) The wheel base units may be identical.

(38) The first and second rescue vehicles may comprise communication means for synchronous operation. This may further increase the chances of a successful lifting operation by preventing jamming which may occur during un-even lifting.
(39) The communication means may enable communication between the first and second rescue vehicles. Such communication may be internal or direct communication between the rescue vehicles. The means of internal or direct communication can be IR, wireless (WiFi), light (LiFi), Bluetooth, NFC or similar.
(40) The system may further comprise a control system, and the control system may comprise cooperative communication means configured to communicate with the communication means of the first and second rescue vehicles to operate synchronously.
(41) The first or second rescue vehicle may be a master rescue vehicle and the other of said first or second rescue vehicle may be a slave rescue vehicle which is at least partly operated by instructions from the master rescue vehicle. The master/slave operation may be limited to the lifting operation, while normal operation of the rescue vehicle(s) in terms of horizontal movement on the rail system may be under control of the control system operating the container handling vehicles. When entering the rail system, the rescue vehicles may be added to the control system operating the remotely operated vehicles such that they are operated as a standard container handling vehicle reducing the probability of collision with the container handling vehicles. Once two rescue vehicles have lifted a malfunctioning container handling vehicle, the control system know how many cell spaces the two rescue vehicles and the malfunctioning container handling vehicle requires, and will take this into consideration when determining what path to use for transporting the malfunctioning vehicle to e.g. a service area.
(42) The rail system may be at a top level of a storage and retrieval system.
(43) The rail system may be a delivery rail system.
(44) The rescue vehicles may comprise two set of wheels for movement in the X and Y directions along the rail system.
(45) The lifting device may comprise an actuator configured to raise and lower the malfunctioning vehicle relative the rail system.
(46) The lifting device may be configured to only move up and down in the Z-direction (i.e. be raised and lowered). This may be achieved using a lifting device in the form of a linear actuator or similar.
(47) However, alternatively, the lifting device may be configured to, in addition to be moved up and down in the Z-direction, also be configured for lateral movement in the X-direction and/or the Y-direction. The latter being advantageous in operations where a larger contact area between the lifting device and the malfunctioning container handling vehicle is required.
(48) At least one of the first and/or second rescue vehicles may comprise at least one rotary drive to winch up a lifting frame and or a track shift motor of the malfunctioning container handling vehicle. This render possible manual/mechanical oversteering of any stuck lifting device or gripper and or set of wheels of the container handling vehicle.
(49) If a container handling vehicle malfunctions with the lifting device/grippers in a lower position, the rescue vehicle may comprise rotary drive for connection to the malfunction vehicle and raise the lifting device/grippers before the malfunctioning vehicle is transported off the rail system.
(50) The rotary drive may be a pipe motor which is used to override the lifting frame/grippers motor and/or track shift motor. One or more cameras arranged on the same or another rescue vehicle may be arranged to monitor the process and provide useful information in terms of aligning the rotary drive relative the complementary socket on the malfunctioning vehicle. In one aspect, the camera may be, or form part of, the communication means between the first and second rescue vehicles.
(51) The at least one rotary drive may be pivotable, and an axis of the rotary drive may be

configured to be pivoted between a stowed vertical position during movement of the first or second rescue vehicle on the rail system, and may further be configured to be pivoted to a deployed horizontal position for winching up the lifting frame and or the track shift motor of a malfunctioning container handling vehicle.
(52) In particular, the rotary drive may be pivoted between its stowed position to the deployed horizontal position as the rescue vehicle is moved towards the malfunctioning vehicle so that the drive shaft can protrude to engage the socket in the malfunctioning vehicle as it approaches.
(53) The rotary drive for the lifting frame may be arranged in an upper part of the rescue vehicle and, when the rotary drive is in the deployed horizontal position, the rotary drive can be supported by an actuator.
(54) Alternatively, the rotary drive can be connected to a linear actuator for movement between the stowed position and the deployed position. When in the stowed position, the perimeter of the rotary drive may be arranged within a horizontal perimeter of the wheel base unit, and when in the deployed position, at least a portion of the rotary drive may extend beyond the perimeter of the wheel base unit. In other words, when in the stowed position, the rescue vehicle may have a footprint equal to or less than a grid cell of the rail system and in the deployed position, the rotary drive may extend into a neighboring grid cell upon actuation of the linear actuator. This provides for the possibility that the service vehicle does not occupy more than one cell when moving on the rail system.
(55) It is further described a rescue vehicle for retrieving a malfunctioning vehicle from a rail system of an automated storage and retrieval system, the rail system comprising a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction perpendicular to the X-direction, a plurality of remotely operated vehicles configured to move in the X and Y-directions on the tracks of the rail system, wherein the rescue vehicle comprises a wheel base unit configured to run on tracks of the rail system, the wheel base unit providing a mobile platform corresponding in area to a single grid cell for a vehicle rescue module mounted thereon, the vehicle rescue module being orientated in a first direction of the rail system;
(56) wherein the vehicle rescue module comprises a lip on at least one of the sides of the module arranged at a level above the level of the wheel base unit.
(57) The rescue vehicle may further comprise a rotary drive to winch up a lifting frame and or a track shift motor of a malfunctioning container handling vehicle.
(58) It is further described an automated storage and retrieval system, comprising: a framework structure comprising upright members, horizontal members and a storage volume comprising storage columns arranged in rows between the upright members and the horizontal members, a plurality of storage containers stacked one on top of one another to form stacks, a rail system comprising a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction perpendicular to the X-direction, a plurality of remotely operated vehicles configured to move in the X and Y-directions on the tracks of the rail system, a rescue system as defined above.
(59) It is further described a method of retrieving a malfunctioning container handling vehicle from a rail system with perpendicular tracks in X and Y direction and wherein a plurality of remotely operated vehicles are arranged on the rail system, and wherein each of the vehicles comprises a vehicle body and side portions, and wherein at least two opposite side portions on each vehicle comprises a recess, wherein the method comprises: determining an anomaly in an operation condition of a vehicle on the rail system, registering the vehicle with the anomalous operational condition as a malfunctioning vehicle, registering a position of the malfunctioning vehicle relative to the supporting rail system, operating a first rescue vehicle configured to run on the tracks of the rail system, the first rescue vehicle being provided with a lifting device on one side of the vehicle, the lifting device facing for engagement in a first X-direction; operating a second rescue vehicle configured to run on the tracks of the rail system, the second rescue vehicle being provided with a

lifting device on an opposite side of the vehicle, the lifting device facing for engagement in a second X-direction opposite to the first; operating the first and second rescue vehicles in tandem by position the first and second rescue vehicles on opposite sides of the malfunctioning vehicle, engaging the respective lifting devices of the first and second rescue vehicles with the opposite sides of the malfunctioning vehicle, operating the lifting devices simultaneously so as to lift the malfunctioning vehicle off the rail system and transport the malfunctioning vehicle.

(60) The malfunctioning container handling vehicle may comprise a lifting device for lifting and lowering storage containers from below, or the malfunctioning container handling vehicle may be a delivery vehicle configured to receive storage containers from above.

(61) It is further described a method of retrieving a malfunctioning container handling vehicle from a rail system with perpendicular tracks in X and Y direction and wherein a plurality of remotely operated vehicles are arranged on the rail system, and wherein each of the vehicles comprises a vehicle body and side portions, and wherein at least two opposite side portions on each vehicle comprises a recess, wherein the method comprises: determining an anomaly in an operation condition of a vehicle on the rail system, registering the vehicle with the anomalous operational condition as a malfunctioning vehicle, registering a position of the malfunctioning vehicle relative to the supporting rail system, operating a first rescue vehicle comprising a first wheel base unit configured to run on the tracks of the rail system, the first wheel base unit providing a mobile platform corresponding in area to a single grid cell for a first vehicle rescue module mounted thereon, the first vehicle rescue module being orientated in a first direction of the rail system; operating a second rescue vehicle comprising a second wheel base unit configured to run on the tracks of the rail system, the second wheel base unit providing a mobile platform corresponding in area to a single grid cell for a second vehicle rescue module mounted thereon, the second vehicle rescue module being orientated in a second direction of the rail system opposite to the first direction; operating the first and second rescue vehicles in tandem to perform rescue operations on a malfunctioning vehicle of the plurality of remotely operated vehicles, engaging the lifting devices on opposite sides of the malfunctioning vehicle, lifting the malfunctioning vehicle off the rail system.

(62) The method may further comprise, prior to the step of engaging the lifting devices on opposite sides of the malfunctioning vehicle, a step of operating a rotary drive to winch up a lifting frame and or a track shift motor of a malfunctioning container handling vehicle.

(63) It is further described a rescue system for retrieving a malfunctioning vehicle from a rail system with perpendicular tracks in X and Y direction and wherein a plurality of remotely operated vehicles are configured to move laterally on the rail system, wherein the rescue system comprises: a first rescue vehicle with a vehicle body, the first rescue vehicle comprising a lifting device configured to be raised and lowered relative an underlying rail system, a second rescue vehicle with a vehicle body, the second rescue vehicle comprising a lifting device configured to be raised and lowered relative the underlying rail system, wherein the first rescue vehicle is configured to extend beyond a perimeter of a first side of the malfunctioning vehicle, and the second rescue vehicle is configured to extend beyond a perimeter of an opposite second side of the malfunctioning vehicle, and wherein, when the first and second rescue vehicles extend beyond the perimeter of the malfunctioning vehicle, the first and second lifting devices are configured to be operated simultaneous such that synchronous operation of the lifting devices lift the malfunctioning vehicle off the rail system.

(64) In all of the disclosed examples, utilizing two rescue vehicles operating in common will require smaller lifting motors than when lifting a container handling vehicle using only one rescue vehicle where one single motor needs to be capable of lifting off the malfunctioning vehicle off the rail system.

(65) Furthermore, the solution provides for more stable transport of malfunctioning vehicle when compared to when using a single rescue vehicle, because the solution gives an advantageous center

of gravity.
(66) Another advantage of the solution is a larger flexibility in accessing areas on the rail system compared to when using smaller rescue vehicles.
(67) In order for the lifting device to support malfunctioning container handling vehicles without moving the lifting device horizontally relative the rescue vehicle during lifting operations, a part of the lifting device such as a lip or similar, may extend into a neighboring cell when the rescue vehicle is positioned in center of a cell. In order for the rescue vehicles to pass by container handling vehicles in neighboring cells, thereby occupying as little space on the rail system as possible, the container handling vehicles preferably have recesses on two or all sides where a neighboring rescue vehicle can pass, while delivery vehicles and so-called single cell robots may have recesses on all sides. The recesses may extend along the whole length of each side of the container handling vehicles. In addition, the recess may have a sufficient extension in the Z direction to take into account different height of the container handling vehicle dependent on which of the set of wheels that are in contact with the rail system.
(68) These can be run on a separate control system, but it would have been simplest and run on the same system as the rest of the vehicles.
(69) It is important to note that the (single-cell) drones can be lifted from both the short side (Y) and the long side (X).
(70) In order to be able to engage with the malfunctioning container handling vehicle, the container handling vehicles may have at least one recess which is complementary shaped relative the lifting plate of the rescue vehicle to be lifted. However, as an alternative, the container handling vehicle may be lifted off the rail system by being sandwiched between two rescue vehicles from opposite sides and then lifted off the rail system, or by using magnets or similar. For example, the rescue vehicles may be fitted with a vertically moveable surface, e.g., a conveyor device or similar—the two rescue vehicles could then push together to hold the malfunctioning vehicle sandwiched between them.
(71) The recess may be at the interface where a carrier module for a container support on a delivery vehicle or a container lifting module is mounted on to a wheel base unit.
(72) Identical rescue vehicles may be used regardless of operating on a top level of a storage and retrieval system or on a delivery rail system, or on a single/single, single/double or double/double track. This provides for large flexibility as the rescue vehicle is the same either in retrieving container handling vehicles with lifting device/grippers or delivery vehicles. For example, using identical rescue vehicles, with opposite orientation, such as one lifting device east-facing and one lifting device west-facing, will provide for the possibility of retrieving both north-facing and south-facing malfunctioning vehicles.
(73) The rescue vehicle may further be provided with visual inspection means such as to perform visual inspection on the rail system or control or check vehicles that have problems on the rail system. The visual inspection means may comprise one or more cameras.
(74) Following drawings are appended to facilitate the understanding of the invention. The drawings show embodiments of the invention, which will now be described by way of example only, where:

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a framework structure of a prior art automated storage and retrieval system;
(2) FIG. **2** is a perspective view of a prior art container handling vehicle having a centrally arranged cavity for carrying storage containers therein;

(3) FIG. **3**A is a perspective view of a prior art container handling vehicle having a cantilever for carrying storage containers underneath;
(4) FIGS. **3**B and **3**C are perspective views of an exemplary automated storage and retrieval system according to the invention, where FIG. **3**B shows a part of the system having a delivery rail system with container delivery vehicles operating below the rail system of container handling vehicles and FIG. **3**C shows an example of a container delivery vehicle having a storage container stored within;
(5) FIGS. **4**A, **4**B and **4**C are perspective views of a rescue vehicle seen from different sides, where the rescue vehicle comprises a pivotable actuator for moving the rotary drive;
(6) FIGS. **5**A and **5**B show an example of a rescue vehicle comprising a linear actuator for moving the rotary drive;
(7) FIGS. **6**A-**6**E show step-by-step two rescue vehicles of FIGS. **4**A-**4**C when they are rescuing a container handling vehicle with a cantilever construction;
(8) FIGS. **7**A-**7**E show step-by-step two rescue vehicles of FIGS. **4**A-**4**C when they are rescuing a container handling vehicle in the form of a delivery vehicle configured to receive storage containers from above;
(9) FIGS. **8**A and **8**B show an exemplary wheel base unit;
(10) FIGS. **9**A-**9**C show an example of the rescue vehicle of FIGS. **4**A-C illustrating step by step the moving of the rotary drive from the stowed vertical to position to the deployed position using a pivotable actuator for manipulating the track shift motor of a container handling vehicle;
(11) FIGS. **10**A and **10**B show an example of the rescue vehicle of FIGS. **9**A-**9**C for manipulating a lifting frame motor of a container handling vehicle;

DETAILED DESCRIPTION OF THE INVENTION

(12) In the following, embodiments of the invention will be discussed in more detail with reference to the appended drawings. It should be understood, however, that the drawings are not intended to limit the invention to the subject-matter depicted in the drawings.
(13) The framework structure **100** of the automated storage and retrieval system **1** is constructed in accordance with the prior art framework structure **100** described above in connection with FIGS. **1-3**, i.e. a number of upright members **102** and a number of horizontal members **103**, which are supported by the upright members **102**, and further that the framework structure **100** comprises a first, upper rail system **108** in the X direction and Y direction.
(14) The framework structure **100** further comprises storage compartments in the form of storage columns **105** provided between the members **102**, **103**, where storage containers **106** are stackable in stacks **107** within the storage columns **105**.
(15) The framework structure **100** can be of any size. In particular it is understood that the framework structure can be considerably wider and/or longer and/or deeper than disclosed in FIG. **1**. For example, the framework structure **100** may have a horizontal extent of more than 700×700 columns and a storage depth of more than twelve containers.
(16) A different automated storage and retrieval system **1** is shown in part in FIG. **3**B. The upright members **102** constitute part of a framework structure **100** onto which a transport rail system **108** with a plurality of container handling vehicles **201**,**301** are operating.
(17) Below this transport rail system **108**, near the floor level, another framework structure **300** is shown which partly extends below some of the storage columns **105** of the framework structure **100**. As for the other framework structure **100**, a plurality of vehicles **340** may operate on a rail system **308** comprising a first set of parallel rails **310** directed in a first direction X and a second set of parallel rails **311** directed in a second direction Y perpendicular to the first direction X, thereby forming a grid pattern in the horizontal plane P.sub.L, comprising a plurality of rectangular and uniform grid locations or grid cells **322**. Each grid cell of this lower rail system **308** comprises a grid opening **315** being delimited by a pair of neighboring rails **310***a*,**310***b* of the first set of rails **310** and a pair of neighboring rails **311***a*,**311***b* of the second set of rails **311**.
(18) The part of the lower rail system **308** that extends below the storage columns **105** are aligned

such that its grid cells **322** are in the horizontal plane P.sub.L, coincident with the grid cells **122** of the upper rail system **108** in the horizontal plane P.
(19) Hence, with this particular alignment of the two rail systems **108**,**308**, a storage container **106** being lowered down into a storage column **105** by a container handling vehicle **250** can be received by a delivery vehicle **340** configured to run on the rail system **308** and to receive storage containers **106** down from the storage column **105**. In other words, the delivery vehicle **340** is configured to receive storage containers **106** from above, preferably directly from the container handling vehicle **201**,**301**.
(20) FIG. **3**C shows an example of such a delivery vehicle **340** comprising a wheel assembly **351** similar to the wheel assembly **251** described for the prior art container handling vehicle **250** and a storage container support **352** for receiving and supporting a storage container **106** delivered by an above container handling vehicle **201**,**301**.
(21) After having received a storage container **106**, the delivery vehicle **340** may drive to an access station adjacent to the rail system **308** (not shown) for delivery of the storage container **106** for further handling and shipping.
(22) FIGS. **4**A, **4**B and **4**C are perspective views of a rescue vehicle **40** seen from different sides. The rescue vehicle **40** comprises a wheel base **2** and a rescue module **48** mounted thereon. The lifting device **41** is disclosed with a vertical plate **42** with a lip **43** extending therefrom. In the disclosed example, the lifting device **41** is on a short side of the rescue vehicle **40** and the lip **43** extends along the whole side and further around a corner of the rescue vehicle **40** and at least partly along an adjacent long side. The part of the lip **43** on the long side of the rescue vehicle may extend shorter and or along the whole length of long side. In one example, there may be a continuous lip **43** around the whole rescue vehicle **40**, i.e. there may be a lip **43** on all sides of the rescue vehicle **40**.
(23) The rescue vehicle **40** is disclosed with visual inspection means **44** such as to perform visual inspection on the rail system **108**,**308** or control or check vehicles that have problems on the rail system **108**, **308**, and or to monitor or assist in rescuing operations. The visual inspection means **44** may comprise one or more cameras.
(24) The rescue vehicle is disclosed with two pivotable actuators **45**′,**45**″ for moving two rotary drives **49**′,**49**″, respectively. The rotary drive denoted **49**′ is for manipulating the track shift motor of a malfunctioning container handling vehicle, whereas the rotary drive denoted **49**″ is for manipulating a lifting frame/gripper motor of a malfunctioning container handling vehicle to rotate the lifting frame motor and any carried storage container **106** up and above the top of the rail system **108**,**308** such that the malfunctioning vehicle can be transported across the rail system **108**,**308**. In all FIGS. **4**A-**4**C, the rotary drives **49**′,**49**″ are in the stowed vertical position. Furthermore, as disclosed in all FIGS. **4**A-**4**C, the footprint of the rescue vehicle **40** is equal to or less than a grid of the underlying rail system **108**,**308**.
(25) FIGS. **5**A and **5**B show an example of a rescue vehicle **40** comprising a wheel base unit **2** and a vehicle rescue module **48** mounted thereon. Instead of the pivotable actuator of the rescue vehicle **40** in FIGS. **4**A-**4**C, the rescue vehicle **20** comprises a linear actuator **46** for moving the rotary drive **49**‴. The linear actuator **46** is arranged for moving the rotary drive **49**‴ between the stowed position (FIG. **5**A) and the deployed position (FIG. **5**B). As shown in FIG. **5**A, when in the stowed position, the perimeter of the rotary drive **49**‴ is arranged within a horizontal perimeter of the wheel base unit **2** of the rescue vehicle **40**, and when in the deployed position, at least a portion of the rotary drive **49**‴ may extend beyond the perimeter of the wheel base unit **2**. The other components of the rescue vehicle **40** may be similar to the ones described in relation to FIGS. **4**A-**4**C.
(26) FIGS. **6**A-**6**E show step-by-step two rescue vehicles of FIGS. **4**A-**4**C when they are rescuing a container handling vehicle **240** with a cantilever construction (i.e. a prior art container handling vehicle **301** as shown in FIG. **3**A).

(27) In FIG. **6**A, the container handling vehicle **240** has malfunctioned with the lifting frame carrying a storage container **106** in an upper position (i.e. a lowermost part of the storage container **106** is above the underlying rail system **108**,**308** such that the container handling vehicle can be transported on the rail system **108**, **308**). The first rescue vehicle **40** comprises a lifting device **41** and is facing for engagement in a first direction, whereas the second rescue vehicle **40** comprises a lifting device on an opposite side of the vehicle relative the first rescue vehicle **40**, the lifting device **41** facing for engagement in a second direction opposite to the first direction. In FIGS. **6**A and **6**B both of the rescue vehicles **40** are positioned in a distance away from the malfunctioning container handling vehicle **240**.
(28) In FIG. **6**C, one of the rescue vehicles **40** has positioned itself in a neighboring cell to the malfunctioning container handling vehicle **240**, on one of the opposite sides of the container handling vehicle **240** which comprise a recess **50** for engagement with the lifting device **40** of the rescue vehicle **40**. In the disclosed example, the lifting device **41** comprises a vertical plate **43** with a lip **43** for engagement with the recess **50**.
(29) In FIG. **6**D, the second rescue vehicle **40** has positioned itself on the opposite side of the malfunctioning container handling vehicle **240** and the lip **43** of the lifting device **41** has engaged with the recess **50**. As is further seen in FIG. **6**D, the first and second rescue vehicles **40** have lifted the malfunctioning container handling vehicle **240** off the rail system **108**,**308** by operating the lifting devices **41** simultaneously (i.e. in tandem). The malfunctioning container handling vehicle **240** may be transported off the rail system **108**,**308** to a dedicated area, such that a service area or similar. FIG. **6**E is an opposite view of FIG. **6**D.
(30) FIGS. **7**A-**7**E show step-by-step two rescue vehicles of FIGS. **4**A-**4**C when they are rescuing a malfunctioning container handling vehicle in the form of a delivery vehicle **340** (see e.g. FIG. **3**C). Similar rescue vehicles **40** as the ones disclosed in FIGS. **6**A-**6**E are used when rescuing the delivery vehicle **340**. The delivery vehicle **340** comprises a recess **50** around the whole circumference of the delivery vehicle **340** for engagement with the lifting device **40** of the rescue vehicle **40**. The recess **340** may be arranged between a wheel base unit **2** of the delivery vehicle **340** and a container supporting unit **3** of the delivery vehicle **340**.
(31) Referring to FIG. **7**A, one of the rescue vehicles **40** has positioned itself in a neighboring cell to the malfunctioning container handling vehicle **340**, on one of the opposite sides of the malfunctioning delivery vehicle **340**. The other rescue vehicle **40** is arranged in a distance from the malfunctioning delivery vehicle **340**.
(32) Referring to FIGS. **7**B and **7**C, the other of the rescue vehicles **40** has positioned itself on an opposite side of the malfunctioning delivery vehicle **340**. FIG. **7**C is an opposite view of FIG. **7**B.
(33) Referring to FIGS. **7**D and **7**E, the first and second rescue vehicles **40** have lifted the malfunctioning delivery vehicle off the rail system **108**,**308** by operating the lifting devices **41** simultaneously (i.e. in tandem). The malfunctioning delivery vehicle **340** may be transported off the rail system **108**,**308** to a dedicated area, such that a service area or similar. FIG. **7**E is a perspective side view of FIG. **7**D.
(34) An exemplary combined wheel base unit for the rescue vehicles **40** and the delivery vehicles **340** is shown in FIGS. **8**A and **8**B. The wheel base unit **2** features a wheel arrangement **32***a*,**32***b* having a first set of wheels **32***a* for movement in a first direction upon a rail system (i.e. any of the top rail system **108** and the delivery rail system **308**) and a second set of wheels **32***b* for movement in a second direction perpendicular to the first direction. Each set of wheels comprises two pairs of wheels arranged on opposite sides of the wheel base unit **2**. To change the direction in which the wheel base unit may travel upon the rail system, one of the sets of wheels **32***b* is connected to a wheel displacement assembly **7**. The wheel displacement assembly is able to lift and lower the connected set of wheels **32***b* relative to the other set of wheels **32***a* such that only the set of wheels travelling in a desired direction is in contact with the rail system. The wheel displacement assembly **7** is driven by an electric motor **8**. Further, two electric motors **4**,**4′**, powered by a rechargeable

battery **6**, are connected to the set of wheels **32***a*,**32***b* to move the wheel base unit in the desired direction.
(35) Further referring to FIGS. **8**A and **8**B, the horizontal periphery of the wheel base unit **2** is dimensioned to fit within the horizontal area defined by a grid cell, such that two wheel base units may pass each other on any adjacent grid cells of the rail system **108**, **308**. In other words, the wheel base unit **2** may have a footprint, i.e. an extent in the X and Y directions, which is generally equal to the horizontal area of a grid cell, i.e. the extent of a grid cell in the X and Y directions, e.g. as is described in WO2015/193278A1, the contents of which are incorporated herein by reference. The wheel base unit **2** has a top panel/flange **9** (i.e. an upper surface) configured as a connecting interface for connection to a connecting interface of a selected vehicle rescue module **48** or container supporting unit **3**. The top panel **9** have a centre opening **20** and features multiple through-holes **10** (i.e. connecting elements) suitable for a bolt **11** connection via corresponding through-holes **10**′ in the connecting interface of a vehicle rescue module **48** or container supporting unit **3**. In other embodiments, the connecting elements of the top panel **9** may for instance be threaded pins for interaction with the through-holes **10**′ of the connecting interface of the vehicle rescue module **48** or container supporting unit **3**, or vice versa. The presence of a centre opening **20** is advantageous as it provides access to internal components of the wheel base unit, such as the rechargeable battery **6** and an electronic control system **21**. The access allows the rechargeable battery **6** and the electronic control system **21** to be easily connected to a rescue module connected to the wheel base unit **2**, thus the vehicle rescue module **48** nor container supporting unit **3** is not required to have its own dedicated power source and/or control system.
(36) FIGS. **9**A-**9**C show an example of the rescue vehicle **40** of FIGS. **4**A-C illustrating step by step the moving of the rotary drive **49**′ from the stowed vertical to the deployed horizontal position using a pivotable actuator **45**′ for manipulating the track shift motor of a container handling vehicle **240**. FIGS. **9**A and **9**B are two different perspective views of a rescue vehicle **40** where the rotary drive **49**′ for manipulating track shift has been pivoted to a deployed horizontal position using a pivotable actuator **45**′. In FIG. **9**C the rotary drive **49**′ is connected to the track shift of the malfunctioning container handling vehicle **240** and can manipulate the track shift motor. The other rotary drive **49**″ for manipulating the lifting frame/gripper motor of a malfunctioning vehicle **240** is in the stowed vertical position in all FIGS. **9**A-**9**C.
(37) FIGS. **10**A and **10**B show an example of the rescue vehicle **40** of FIGS. **9**A-**9**C for manipulating a lifting frame motor of a container handling vehicle, where in FIG. **10**A the rotary drive **49**″ for manipulating the lifting frame motor is in a deployed horizontal position actuated by a pivotable actuator **4**″ but not connected to a container handling vehicle **240**, whereas in FIG. **10**B the rotary drive **49**″ is connected to an interface for winching up the lifting frame motor of the container handling vehicle **240** such that the lifting frame (possibly carrying a storage container **106**) can be lifted up and above the rail system making it possible to transport the container handling vehicle **240**. When the rotary drive **49**″ is in the stowed position, the rescue vehicle **40** has a footprint equal to or less than a grid cell of the rail system and in the deployed position, the rotary drive **49**″ extend into a neighboring grid cell by being pivoted from the deployed vertical position to the deployed horizontal position. This provides for the possibility that the service vehicle **40** does not occupy more than one cell when moving on the rail system.
(38) In the preceding description, various aspects of the delivery vehicle and the automated storage and retrieval system according to the invention have been described with reference to the illustrative embodiment. For purposes of explanation, specific numbers, systems and configurations were set forth in order to provide a thorough understanding of the system and its workings. However, this description is not intended to be construed in a limiting sense. Various modifications and variations of the illustrative embodiment, as well as other embodiments of the system, which are apparent to persons skilled in the art to which the disclosed subject matter pertains, are deemed to lie within the scope of the present invention.

LIST OF REFERENCE NUMBERS

(39) TABLE-US-00001 1 Prior art storage and retrieval system 2 Wheel base unit 3 Container supporting unit 4, 4′ Electric motor 6 Rechargeable battery 7 Wheel displacement assembly 8 Electric motor for wheel displacement assembly 9 Top panel/flange 10 Through-holes 11 bolt 20 Centre opening 21 Electronic control system 32a, b Wheel arrangement on wheel base unit, first and second set of wheels 40 Rescue vehicle 41 Lifting device 42 Vertical plate 43 lip 44 Visual inspection means/camera 45′, 45″ Pivotable actuator 46 Linear actuator 48 Rescue module 49′, 49″, 49′′′ Rotary drive 50 Recess 100 Framework structure 102 Upright members of framework structure 103 Horizontal members of framework structure 104 Storage grid 105 Storage column 106 Storage container 106′ Particular position of storage container 107 Stack 108 Rail system 110 Parallel rails in first direction (X) 110a First rail in first direction (X) 110b Second rail in first direction (X) 111 Parallel rail in second direction (Y) 111a First rail of second direction (Y) 111b Second rail of second direction (Y) 112 Access opening 119 First port column 120 Second port column 201 Prior art storage container vehicle 201a Vehicle body of the storage container vehicle 101 201b Drive means/wheel arrangement, first direction (X) 201c Drive means/wheel arrangement, second direction (Y) 240 Malfunctioning container handling vehicle 300 Delivery framework structure 301 Prior art cantilever storage container vehicle 301a Vehicle body of the storage container vehicle 101 301b Drive means in first direction (X) 301c Drive means in second direction (Y) 308 Delivery rail system 310 First set of parallel rails in first direction (X) on delivery rail system 311 Second set of parallel rails in second direction (Y) on delivery rail system 315 Grid opening in delivery rail system 322 Grid cell of delivery rail system 340 Malfunctioning container delivery vehicle 351 Wheel assembly for container delivery vehicle 352 Storage container support 500 Control system X First direction Y Second direction Z Third direction

Claims

1. A rescue system for retrieving a malfunctioning vehicle from a rail system of an automated storage and retrieval system, the rail system comprising a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction perpendicular to the X-direction, and a plurality of remotely operated vehicles configured to move in the X-direction and the Y-direction on the tracks of the rail system, wherein the rescue system comprises: a first rescue vehicle configured to run on the tracks of the rail system, the first rescue vehicle comprising a lifting device on one side of the first rescue vehicle, the lifting device configured to engage in a first X-direction, and a part of the lifting device extending into a neighboring cell when the first rescue vehicle is positioned in a center of a grid cell; and a second rescue vehicle configured to run on the tracks of the rail system, the second rescue vehicle comprising a lifting device on an opposite side of the second rescue vehicle relative to the one side of the first rescue vehicle, the lifting device configured to engage in a second X-direction opposite to the first X-direction, and a part of the lifting device extending into a neighboring cell when the second rescue vehicle is positioned in a center of a grid cell; wherein the first rescue vehicle and the second rescue vehicle are configured to work in tandem so that when one of the plurality of remotely operated vehicles malfunctions, the first rescue vehicle and the second rescue vehicle position themselves on the rail system on opposite sides of the malfunctioning vehicle to engage their respective lifting devices with the opposite sides of the malfunctioning vehicle and operate their lifting devices simultaneously so as to lift the malfunctioning vehicle off the rail system and transport the malfunctioning vehicle.
2. The rescue system of claim 1, wherein each respective lifting device comprises a vertical plate with a lip extending therefrom, and wherein the part of each respective lifting device is the corresponding lip.

3. The rescue system of claim 1, wherein the rescue system further comprises a control system, and wherein the control system comprises cooperative communication means configured to communicate with the communication means of the first rescue vehicle and the second rescue vehicle to operate synchronously.
4. The rescue system of claim 1, wherein the rail system is at a top level of the automated storage and retrieval system.
5. The rescue system of claim 1, wherein the rail system is a delivery rail system.
6. The rescue system of claim 1, wherein each of the first rescue vehicle and the second rescue vehicle comprises two set of wheels for movement in the X-direction and the Y-direction along the rail system.
7. The rescue system of claim 1, wherein the lifting device comprises an actuator configured to raise and lower the malfunctioning vehicle relative the rail system.
8. The rescue system of claim 1, wherein the first rescue vehicle and the second rescue vehicle each comprise communication means for synchronous operation.
9. The rescue system of claim 8, wherein the communication means enables communication between the first rescue vehicle and the second rescue vehicle.
10. The rescue system of claim 9, wherein the first rescue vehicle is a master rescue vehicle and the second rescue vehicle is a slave rescue vehicle configured to be at least partly operated by instructions from the master rescue vehicle; or wherein the second rescue vehicle is a master rescue vehicle and the first rescue vehicle is a slave rescue vehicle configured to be at least partly operated by instructions from the master rescue vehicle.
11. The rescue system of claim 1, wherein at least one of the first rescue vehicle or the second rescue vehicle comprises at least one rotary drive to winch up one or more of: a lifting frame; and a track shift motor of the malfunctioning vehicle.
12. The rescue system of claim 11, wherein the at least one rotary drive is pivotable, an axis of the rotary drive is configured to be pivoted between a stowed vertical position during movement of at least one of the first rescue vehicle or the second rescue vehicle on the rail system, and the axis of the rotary drive is further configured to be pivoted to a deployed horizontal position for winching up one or more of: a lifting frame; and a track shift motor of the malfunctioning vehicle.
13. The rescue system of claim 12, wherein the rotary drive for the lifting frame is arranged in an upper part of the rescue vehicle, and wherein, when in the deployed horizontal position, the rotary drive is supported by an actuator.
14. The rescue system of claim 12, wherein the rotary drive is connected to a linear actuator for movement between the stowed vertical position and the deployed horizontal position, and wherein, when in the stowed vertical position, a perimeter of the rotary drive is arranged within a horizontal perimeter of a wheel base unit, and wherein, when in the deployed horizontal position, at least a portion of the rotary drive extends beyond a perimeter of a wheel base unit.
15. A rescue system for retrieving a malfunctioning vehicle from a rail system of an automated storage and retrieval system, the rail system comprising a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction perpendicular to the X-direction, the rails defining a plurality of grid cells, and a plurality of remotely operated vehicles configured to move in the X-direction and Y-direction on the tracks of the rail system, wherein the rescue system comprises: a first rescue vehicle comprising a first wheel base unit configured to run on the tracks of the rail system, the first wheel base unit providing a mobile platform corresponding in area to a single grid cell for a first vehicle rescue module mounted thereon, wherein the first vehicle rescue module is orientated in a first direction of the rail system; a second rescue vehicle comprising a second wheel base unit configured to run on the tracks of the rail system, the second wheel base unit providing a mobile platform corresponding in area to a single grid cell for a second vehicle rescue module mounted thereon, wherein the second vehicle rescue module is orientated in a second direction of the rail system opposite to the first direction; wherein the first rescue vehicle

and the second rescue vehicle are configured to work in tandem to perform rescue operations on the malfunctioning vehicle of the plurality of remotely operated vehicles, the first vehicle rescue module and the second vehicle rescue module comprise a lifting device; and wherein a part of the lifting device of the first vehicle rescue module extends into a neighboring cell when the first rescue vehicle is positioned in a center of a grid cell, a part of the lifting device of the second vehicle rescue module extends into a neighboring cell when the second rescue vehicle is positioned in a center of a grid cell, and the first vehicle rescue module and the second vehicle rescue module are arranged to engage opposite sides of the malfunctioning vehicle and lift the malfunctioning vehicle off the rail system using the respective lifting devices orientated in opposite facing directions.

16. The rescue system of claim 15, wherein the first vehicle rescue module and the second vehicle rescue module each comprise a lifting plate with a lip extending at a height above an upper surface of each respective wheel base unit; and wherein the part of each respective lifting device is the corresponding lip.

17. The rescue system of claim 15, wherein each remotely operated vehicle of the plurality of remotely operated vehicles comprises wheel base units providing mobile platforms, each mobile platform corresponding in area to a single grid cell of the rail system, for storage container lifting modules mounted thereon.

18. The rescue system of claim 15, wherein the wheel base units are identical.

19. A rescue vehicle for retrieving a malfunctioning vehicle from a rail system of an automated storage and retrieval system, the rail system comprising a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction perpendicular to the X-direction, a plurality of remotely operated vehicles configured to move in the X-direction and the Y-direction on the tracks of the rail system, wherein the rescue vehicle comprises a wheel base unit configured to run on the tracks of the rail system, the wheel base unit providing a mobile platform corresponding in area to a single grid cell for a vehicle rescue module mounted thereon, the vehicle rescue module being orientated in a first direction of the rail system; wherein the vehicle rescue module comprises a lip on at least one side of the vehicle rescue module arranged at a level above a level of the wheel base unit; and the lip extends into a neighboring cell when the rescue vehicle is positioned in a center of a grid cell.

20. The rescue vehicle of claim 19, further comprising a rotary drive to winch up one or more of: a lifting frame; and a track shift motor of the malfunctioning vehicle.

21. An automated storage and retrieval system, comprising: a framework structure comprising upright members, horizontal members and a storage volume, the storage volume comprising storage columns arranged in rows between the upright members and the horizontal members, a plurality of storage containers stacked one on top of one another to form stacks, a rail system comprising a plurality of rails with tracks extending in an X-direction and a plurality of rails with tracks extending in a Y-direction perpendicular to the X-direction; a plurality of remotely operated vehicles configured to move in the X-direction and the Y-direction on the tracks of the rail system; and a rescue system for retrieving a malfunctioning vehicle from the rail system, the rescue system comprising: a first rescue vehicle configured to run on the tracks of the rail system, the first rescue vehicle comprising a lifting device on one side of the first rescue vehicle, the lifting device facing for engagement in a first X-direction; a second rescue vehicle configured to run on the tracks of the rail system, the second rescue vehicle comprising a lifting device on an opposite side of the second rescue vehicle relative to the one side of the first rescue vehicle, the lifting device configured to engage in a second X-direction opposite to the first X-direction; and wherein the first rescue vehicle and the second rescue vehicle are configured to work in tandem so that when one of the plurality of remotely operated vehicles malfunctions, the first rescue vehicle and the second rescue vehicle position themselves on the rail system on opposite sides of the malfunctioning vehicle to engage their respective lifting devices with the opposite sides of the malfunctioning vehicle and

operate their lifting devices simultaneously so as to lift the malfunctioning vehicle off the rail system and transport the malfunctioning vehicle.

22. A method of retrieving a malfunctioning vehicle from a rail system with perpendicular tracks in an X-direction and a Y-direction, wherein a plurality of remotely operated vehicles are arranged on the rail system, each remotely operated vehicle of the plurality of remotely operated vehicles comprises a vehicle body and side portions, wherein at least two opposite side portions on each remotely operated vehicle of the plurality of remotely operated vehicles comprises a recess, and wherein the method comprises: determining an anomaly in an operational condition of a remotely operated vehicle of the plurality of remotely operated vehicles on the rail system; registering the remotely operated vehicle with the anomalous operational condition as a malfunctioning vehicle; registering a position of the malfunctioning vehicle relative to the rail system; operating a first rescue vehicle configured to run on the perpendicular tracks, the first rescue vehicle provided with a lifting device on one side of the first rescue vehicle, the lifting device configured to engage in a first X-direction, and a part of the lifting device extending into a neighboring cell when the first rescue vehicle is positioned in a center of a grid cell; operating a second rescue vehicle configured to run on the perpendicular tracks of, the second rescue vehicle comprising a lifting device on an opposite side of the second rescue vehicle relative to the one side of the first rescue vehicle, the lifting device configured to engage in a second X-direction opposite to the first X-direction, and a part of the lifting device extending into a neighboring cell when the second rescue vehicle is positioned in a center of a grid cell; operating the first rescue vehicle and the second rescue vehicle in tandem by positioning the first rescue vehicle and the second rescue vehicle on opposite sides of the malfunctioning vehicle; engaging the respective lifting devices with the opposite sides of the malfunctioning vehicle; and operating the lifting devices simultaneously so as to lift the malfunctioning vehicle off the rail system and transport the malfunctioning vehicle.

23. The method of claim 22, wherein the method further comprises, prior to engaging the parts of the respective lifting devices, operating a rotary drive to winch up one or more of: a lifting frame; and a track shift motor of the malfunctioning vehicle.

24. A method of retrieving a malfunctioning vehicle from a rail system with perpendicular tracks in X-direction and Y-direction, wherein a plurality of remotely operated vehicles are arranged on the rail system, each remotely operated vehicle of the plurality of remotely operated vehicles comprises a vehicle body and side portions, wherein at least two opposite side portions on each remotely operated vehicle of the plurality of remotely operated vehicles comprises a recess, and wherein the method comprises: determining an anomaly in an operational condition of a remotely operated vehicle of the plurality of remotely operated vehicles on the rail system; registering the remotely operated vehicle with the anomalous operational condition as a malfunctioning vehicle; registering a position of the malfunctioning vehicle relative to the rail system; operating a first rescue vehicle comprising a first wheel base unit, the first rescue vehicle configured to run on the perpendicular tracks, the first wheel base unit providing a mobile platform corresponding in area to a single grid cell for a first vehicle rescue module mounted thereon, the first vehicle rescue module orientated in a first direction of the rail system; operating a second rescue vehicle comprising a second wheel base unit, the second rescue vehicle configured to run on the perpendicular tracks, the second wheel base unit providing a mobile platform corresponding in area to a single grid cell for a second vehicle rescue module mounted thereon, the second vehicle rescue module orientated in a second direction of the rail system opposite to the first direction; operating the first rescue vehicle and the second rescue vehicle in tandem to perform rescue operations on the malfunctioning vehicle, the first vehicle rescue module and the second vehicle rescue module each comprising a lifting device, wherein a part of the lifting device of the first vehicle rescue module extends into a neighboring cell when the first rescue vehicle is positioned in a center of a grid cell, and wherein a part of the lifting device of the second vehicle rescue module extends into a neighboring cell when the second rescue vehicle is positioned in a center of a grid cell; engaging the parts of the respective lifting

devices that extend into the respective neighboring cells on opposite sides of the malfunctioning vehicle, lifting the malfunctioning vehicle off the rail system.